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System and method for continuous and on-demand analyte monitoring

Abstract

An analyte monitoring system includes an analyte sensor, a transceiver, and a display device. The analyte sensor may include an analyte indicator that exhibits one or more detectable properties based on an amount or concentration of an analyte in proximity to the analyte indicator. The transceiver may receive first sensor data directly from the analyte sensor and may calculate first analyte information using at least the received first sensor data. The display device may receive second sensor data directly from the analyte sensor and calculates second analyte information using at least the received second sensor data. The transceiver may convey the first analyte information, and the display device may receive the first analyte information conveyed by the transceiver and display the first and second analyte information.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a divisional of U.S. application Ser. No. 16/709,140, filed Dec. 10, 2019, which claims the benefit of priority to U.S. Provisional Application Ser. No. 62/777,583, filed on Dec. 10, 2018, which are incorporated herein by reference in their entireties.

BACKGROUND

Field of Invention

(1) The present disclosure relates to an analyte monitoring system and method. More specifically, aspects of the present disclosure relate to an analyte monitoring system for continuous and on-demand analyte monitoring.

Discussion of the Background

(2) Analyte monitoring systems may be used to measure analyte levels, such as analyte concentrations. One type of analyte monitoring system is a continuous glucose monitoring (CGM) system. A CGM system measures glucose levels throughout the day and can be very useful in the management of diabetes.

(3) Some CGM systems include an analyte sensor and a transmitter that are worn by the patient. In some examples, the analyte sensor and the transmitter are worn on either the arm or the stomach of the host, which typically require some type of tape or strap to ensure the analyte sensor and transmitter remain mounted to the skin of the patient. While using a CGM system is helpful for the patient to manage his or her glucose levels, continuously wearing an analyte sensor or a transmitter throughout the day can be cumbersome for the patient, interfering with the patient's daily activities. Accordingly, improved analyte monitoring systems and methods are needed.

SUMMARY

(4) The present invention overcomes the disadvantages of prior systems by providing, among other advantages, more flexibility in choosing when to wear or use the analyte monitoring system while still providing the patient the ability to monitor his or her glucose levels at any time. In some embodiments, the present invention may provide an improved analyte monitoring system that operates under two or more modes, including a continuous glucose monitoring mode and a flash glucose monitoring mode (e.g., on-demand request for sensor data). A user or patient may select the analyte monitoring system to operate under either the continuous glucose monitoring mode or the flash glucose monitoring mode, as desired.

(5) One aspect of the invention may provide an analyte monitoring system including an analyte sensor, a transceiver, and a display device. The analyte sensor may include an analyte indicator that exhibits one or more detectable properties based on an amount or concentration of an analyte in proximity to the analyte indicator. The transceiver may be configured to receive first sensor data directly from the analyte sensor, calculate first analyte information using at least the received first sensor data, and convey the first analyte information. The display device may be configured to receive second sensor data conveyed from the analyte sensor, calculate analyte information using at least the received second sensor data, receive the first analyte information conveyed by the transceiver, and display the first and second analyte information.

(6) In some embodiments, the display device may include a first wireless communication integrated-circuit (IC) and a second wireless communication IC. In some embodiments, the first communication IC may be configured to employ a first standard to communicate wirelessly, the second communication IC may be configured to employ a second standard to communicate wirelessly, and the second standard may be different than the first standard. In some embodiments,

the first standard is a Bluetooth standard, and the second standard is a near field communication (NFC) standard. In some embodiments, the display device may be configured to use the first wireless communication IC to receive the first analyte information conveyed by the transceiver, and the display device may be configured to use the second wireless communication IC to receive the second sensor data directly from the analyte sensor. In some embodiments, the display device may include a third wireless communication IC, and the display device may be configured to use the third wireless communication IC to convey the first and second analyte information over a network to a remote device.

(7) In some embodiments, the transceiver may be worn by a host using the analyte sensor while the transceiver receives the first sensor data directly from the analyte sensor. In some embodiments, the display device may include a graphical user interface and may be configured to generate an alert or alarm on the graphical user interface of the display device. In some embodiments, the transceiver may include a graphical user interface and may be configured to generate an alert or alarm on the graphical user interface of the transceiver. In some embodiments, the display device may be configured to convey the second analyte information to the transceiver, and the transceiver may be configured to receive the second analyte information.

(8) In some embodiments, the display device may be configured to receive one or more calibration points. In some embodiments, the display device may be configured to perform an analyte information calibration using at least the one or more calibration points. In some embodiments, the display device may be configured convey the one or more calibration points, and the transceiver may be configured to receive the one or more calibration points and perform an analyte information calibration using at least the one or more calibration points.

(9) In some embodiments, one or more of the first and second analyte information may include one or more of: (i) an analyte concentration, (ii) a time stamp, and (iii) an analyte concentration trend information. In some embodiments, one or more of the first and second analyte information may include one or more of: (i) an alert, (ii) an alarm, and (iii) a notification.

(10) In some embodiments, the first sensor data may include one or more of: (i) a measurement of the one or more detectable properties and (ii) a temperature measurement.

(11) Another aspect of the present invention may provide a method for using analyte monitoring system. The method may include a step (a) of using a transceiver of an analyte monitoring system to receive first sensor data directly from an analyte sensor of the analyte monitoring system. The method may include a step (b) of using the transceiver to calculate first analyte information using at least the first sensor data. The method may include a step (c) of using the transceiver to convey the first analyte information. The method may include a step (d) of using a display device of the analyte monitoring system to receive the first analyte information conveyed by the transceiver. The method may include a step (e) of using the display device to receive second sensor data directly from the analyte sensor. The method may include a step (f) of using the display device to calculate second analyte information using at least the received second sensor data. The method may include a step (g) of using the display device of using the display device to display the first analyte information and the second analyte information.

(12) In some embodiments, the step (a) may include positioning the transceiver such that the transceiver is operatively linked to the analyte sensor. In some embodiments, positioning the transceiver may include wearing, by a host using the analyte sensor, the transceiver externally on an armband, a wrist band, a waist band, or an adhesive patch. In some embodiments, the step (e) may include positioning the display device such that the display device is operatively linked to analyte sensor. In some embodiments, the step (e) may include removing the transceiver away from the analyte sensor such that the transceiver is not operatively linked to the analyte sensor.

(13) In some embodiments, the display device may include a first wireless communication IC and a second wireless communication IC, and the first communication IC may be configured to employ a first standard to communicate wirelessly, the second communication IC may be configured to

employ a second standard to communicate wirelessly, and the second standard may be different than the first standard. In some embodiments, the first standard may be a Bluetooth standard, and the second standard may be a near field communication (NFC) standard. In some embodiments, the step (c) may include using the first communication IC of the display device to receive the first analyte information conveyed by the transceiver. In some embodiments, the step (e) may include using the second communication IC of the display device to receive the second sensor data directly from the sensor. In some embodiments, the display device may include a third wireless communication IC, and the method may include using the third wireless communication IC to convey the first analyte information and the second analyte information over a network to a remote device.

(14) In some embodiments, one or more of the first and second analyte information may include one or more of: (i) an analyte concentration, (ii) a time stamp, and (iii) analyte concentration trend information. In some embodiments, one or more of the first and second analyte information may include one or more of: (i) an alert, (ii) an alarm, and (iii) a notification. In some embodiments, the first sensor data may include one or more of: (i) a measurement of one or more detectable properties exhibited by an analyte indicator of the analyte sensor based on an amount or concentration of an analyte in proximity to the analyte indicator and (ii) a temperature measurement.

(15) In some embodiments, the display device may include a graphical user interface and may be configured to generate an alert or alarm on the graphical user interface of the display device. In some embodiments, the transceiver may include a graphical user interface and may be configured to generate an alert or alarm on the graphical user interface of the transceiver. In some embodiments, the method may further include using the display device to convey the second analyte information and using the transceiver to receive the second analyte information conveyed by the display device.

(16) In some embodiments, the method may further include using the display device to receive one or more calibration points. In some embodiments, the method may further include using the display device to perform an analyte information calibration using at least the one or more calibration points. In some embodiments, the method may further include using the display device to convey the one or more calibration points, and the method may further include using the transceiver to receive the one or more calibration points conveyed by the display device and perform an analyte information calibration using at least the one or more calibration points.

(17) Another aspect of the present invention may provide an analyte monitoring system comprising an analyte sensor, a transceiver, and a display device. The analyte sensor may include an analyte indicator that exhibits one or more detectable properties based on an amount or concentration of an analyte in proximity to the analyte indicator. The transceiver may be configured to receive first sensor data directly from the analyte sensor and to convey the first sensor data. The display device may be configured to receive the first sensor data conveyed by the transceiver, calculate first analyte information using at least the received first sensor data, receive second sensor data directly from the analyte sensor, calculate second analyte information using at least the received second sensor data, and display the first and second analyte information.

(18) In some embodiments, the display device may include a first wireless communication integrated-circuit (IC) and a second wireless communication IC. In some embodiments, the first communication IC may be configured to employ a first standard to communicate wirelessly, the second communication IC may be configured employ a second standard to communicate wirelessly, and the second standard may be different than the first standard. In some embodiments, the first standard is a Bluetooth standard, and the second standard is a near field communication (NFC) standard. In some embodiments, the display device may be configured to use the first wireless communication IC to receive the first sensor data conveyed by the transceiver, and the display device may be configured to use the second wireless communication IC to receive the

second sensor data directly from the analyte sensor. In some embodiments, the display device may include a third wireless communication IC, and the display device may be configured to use the third wireless communication IC to convey the first and second analyte information over a network to a remote device.

(19) Another aspect of the present invention may provide a method of using an analyte monitoring system. The method may include a step (a) of using a transceiver of an analyte monitoring system to receive first sensor data directly from an analyte sensor of the analyte monitoring system. The method may include a step (b) of using the transceiver to convey the first analyte information. The method may include a step (c) of using a display device of the analyte monitoring system to receive the first sensor data conveyed by the transceiver. The method may include a step (d) of using the display device to calculate first analyte information using at least the first sensor data. The method may include a step (e) of using the display device to receive second sensor data directly from the analyte sensor. The method may include a step (f) of using the display device to calculate second analyte information using at least the received second sensor data. The method may include a step (g) of using the display device to display the first analyte information and the second analyte information.

(20) In some embodiments, the step (a) may include positioning the transceiver such that the transceiver is operatively linked to the analyte sensor. In some embodiments, positioning the transceiver may include wearing, by a host using the analyte sensor, the transceiver externally on an armband, a wrist band, a waist band, or an adhesive patch.

(21) In some embodiments, the step (e) may include positioning the display device such that the display device is operatively linked to analyte sensor. In some embodiments, the step (e) may include using the second communication IC of the display device to receive the second sensor data from the analyte sensor.

(22) In some embodiments, the display device may include a first wireless communication IC and a second wireless communication IC, the first communication IC may be configured to employ a first standard to communicate wirelessly, the second communication IC may be configured to employ a second standard to communicate wirelessly, and the second standard may be different than the first standard. In some embodiments, the first standard may be a Bluetooth standard, and the second standard may be a near field communication (NFC) standard. In some embodiments, the step (b) may include using the first communication IC of the display device to receive the first analyte sensor from the transceiver.

(23) In some embodiments, the display device may include a third wireless communication IC, and the method may further include using the third wireless communication of the display device to convey the first analyte information and the second analyte information over a network to a remote device. In some embodiments, one or more of the first and second analyte information may include one or more of: (i) an analyte concentration, (ii) a time stamp, and (iii) analyte concentration trend information. In some embodiments, the first and second analyte information may include one or more of: (i) an alert, (ii) an alarm, and (iii) a notification. In some embodiments, the first sensor data may include one or more of: (i) a measurement of the one or more detectable properties exhibited by an analyte indicator of the analyte sensor based on an amount or concentration of an analyte in proximity to the analyte indicator and (ii) a temperature measurement.

(24) In some embodiments, the display device may include a graphical user interface and may be configured to generate an alert or alarm on the graphical user interface of the display device. In some embodiments, the transceiver comprises a graphical user interface and is configured to generate an alert or alarm on the graphical user interface of the transceiver. In some embodiments, the method may further include using the display device to convey the second analyte information to the transceiver. In some embodiments, the method may further include using the display device to receive one or more calibration points. In some embodiments, the method may further include using the display device to perform an analyte information calibration using at least the one or

more calibration points.

(25) Another aspect of the invention may provide a display device. The display device may include a first wireless communication integrated circuit (IC) configured to employ a first standard to communicate wirelessly with a transceiver and to receive first analyte information from the transceiver. The display device may include a second wireless communication IC configured to employ a second standard to communicate wirelessly and directly with an analyte sensor and to receive sensor data directly from the analyte sensor, and the second standard may be different than the first standard. The display device may include a graphical user interface. The display device may include a computer including a non-transitory memory and a processor. The computer may be configured to calculate second analyte information using at least the received second sensor data and to display the first and second analyte information using the graphical user interface.

(26) Another aspect of the invention may provide a display device. The display device may include a first wireless communication integrated circuit (IC) configured to employ a first standard to communicate wirelessly with a transceiver and to receive first sensor data from the transceiver. The display device may include a second wireless communication IC configured to employ a second standard to communicate wirelessly and directly with an analyte sensor and to receive second sensor data directly from the analyte sensor, and the second standard may be different than the first standard. The display device may include a graphical user interface. The display device may include a computer including a non-transitory memory and a processor. The display device may be configured to: (i) calculate first analyte information using at least the received first sensor data, (ii) calculate second analyte information using at least the received second sensor data, and (iii) display the first and second analyte information using the graphical user interface.

(27) These and other embodiments encompassed within the systems and methods are described in the detailed description of the invention below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated herein and form part of the specification, illustrate various, non-limiting embodiments of the present invention. In the drawings, like reference numbers indicate identical or functionally similar elements.

(2) FIG. 1 illustrates a schematic view of an analyte monitoring system embodying aspects of the present invention.

(3) FIG. 2 illustrates a schematic view of a sensor and transceiver of an analyte monitoring system embodying aspects of the present invention.

(4) FIG. 3 illustrates a cross-sectional, perspective view of a transceiver embodying aspects of the invention.

(5) FIG. 4 illustrates an exploded, perspective view of a transceiver embodying aspects of the invention.

(6) FIG. 5 illustrates a schematic view illustrating a transceiver embodying aspects of the present invention.

(7) FIG. 6 illustrates a block diagram of a display device of the analyte monitoring system embodying aspects of the present invention.

(8) FIG. 7 illustrates a block diagram of a computer of the display device of the analyte monitoring system according to some embodiments.

(9) FIG. 8 illustrates a non-limiting example of a home screen illustrative display of a medical mobile application embodying aspects of various embodiments of the present invention.

(10) FIG. 9 is a flow chart illustrating a method of using an analyte monitoring system embodying aspects of the present invention.

(11) FIG. **10** is a flow chart illustrating a method of using an analyte monitoring system embodying aspects of the present invention.

(12) FIG. **11** is a flow chart illustrating a method of using a display device embodying aspects of the present invention

(13) FIG. **12** is a flow chart illustrating a method of using a display device embodying aspects of the present invention.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

(14) FIG. **1** is a schematic view of an exemplary analyte monitoring system **50** embodying aspects of the present invention. In some embodiments, the analyte monitoring system **50** may be a continuous analyte monitoring system (e.g., a continuous glucose monitoring system). In some embodiments, the analyte monitoring system **50** may be an on-demand monitoring system. In some embodiments, the analyte monitoring system **50** may include one or more of an analyte sensor **100**, a transceiver **101**, a display device **105**, a personal computer **109**, and a data management system **111** hosted by a remote server or network attached storage hardware.

(15) In some embodiments, the sensor **100** may be small, fully subcutaneously implantable sensor measures analyte (e.g., glucose) concentrations in a medium (e.g., interstitial fluid) of a living animal (e.g., a living human). However, this is not required, and, in some alternative embodiments, the sensor **100** may be a partially implantable (e.g., transcutaneous) sensor or a fully external sensor. In some embodiments, the transceiver **101** may be an externally worn transceiver (e.g., attached via an armband, wristband, waistband, or adhesive patch). In some embodiments, the transceiver **101** may remotely power and/or communicate with the sensor to initiate and receive the measurements (e.g., via near field communication (NFC)). However, this is not required, and, in some alternative embodiments, the transceiver **101** may power and/or communicate with the sensor **100** via one or more wired connections. In some non-limiting embodiments, the transceiver **101** may be a smartphone (e.g., an NFC-enabled smartphone). In some embodiments, the transceiver **101** may communicate information (e.g., one or more analyte concentrations) wirelessly (e.g., via a Bluetooth™ communication standard such as, for example and without limitation Bluetooth Low Energy) to a hand held application running on a display device **105** (e.g., smartphone). In some embodiments, the analyte monitoring system **50** may include a web interface for plotting and sharing of uploaded data.

(16) In some embodiments, as illustrated in FIG. **2**, the transceiver **101** may include an inductor **103**, such as, for example, a coil. The transceiver **101** may generate an electromagnetic wave or electrodynamic field (e.g., by using a coil) to induce a current in an inductor **114** of the sensor **100**, which powers the sensor **100**. The transceiver **101** may also convey data (e.g., commands) to the sensor **100**. For example, in a non-limiting embodiment, the transceiver **101** may convey data by modulating the electromagnetic wave used to power the sensor **100** (e.g., by modulating the current flowing through a coil **103** of the transceiver **101**). The modulation in the electromagnetic wave generated by the transceiver **101** may be detected/extracted by the sensor **100**. Moreover, the transceiver **101** may receive sensor data (e.g., measurement information) directly from the sensor **100**. For example, in a non-limiting embodiment, the transceiver **101** may receive sensor data by detecting modulations in the electromagnetic wave generated by the sensor **100**, e.g., by detecting modulations in the current flowing through the coil **103** of the transceiver **101**.

(17) The inductor **103** of the transceiver **101** and the inductor **114** of the sensor **100** may be in any configuration that permits adequate field strength to be achieved when the two inductors are brought within adequate physical proximity.

(18) In some non-limiting embodiments, as illustrated in FIG. **2**, the sensor **100** may be encased in a sensor housing **102** (i.e., body, shell, capsule, or encasement), which may be rigid and biocompatible. The sensor **100** may include an analyte indicator **106**, such as, for example, a polymer graft coated, diffused, adhered, or embedded on or in at least a portion of the exterior surface of the sensor housing **102**. The analyte indicator **106** (e.g., polymer graft) of the sensor **100**

may include indicator molecules **104** (e.g., fluorescent indicator molecules) exhibiting one or more detectable properties (e.g., optical properties) based on the amount or concentration of the analyte in proximity to the analyte indicator **106**. In some embodiments, the sensor **100** may include a light source **108** that emits excitation light **329** over a range of wavelengths that interact with the indicator molecules **104**. The sensor **100** may also include one or more photodetectors **224**, **226** (e.g., photodiodes, phototransistors, photoresistors, or other photosensitive elements). The one or more photodetectors (e.g., photodetector **224**) may be sensitive to emission light **331** (e.g., fluorescent light) emitted by the indicator molecules **104** such that a signal generated by a photodetector (e.g., photodetector **224**) in response thereto that is indicative of the level of emission light **331** of the indicator molecules and, thus, the amount of analyte of interest (e.g., glucose). In some non-limiting embodiments, one or more of the photodetectors (e.g., photodetector **226**) may be sensitive to excitation light **329** that is reflected from the analyte indicator **106** as reflection light **333**. In some non-limiting embodiments, one or more of the photodetectors may be covered by one or more filters that allow only a certain subset of wavelengths of light to pass through (e.g., a subset of wavelengths corresponding to emission light **331** or a subset of wavelengths corresponding to reflection light **333**) and reflect the remaining wavelengths. In some non-limiting embodiments, the sensor **100** may include a temperature transducer **670**. In some non-limiting embodiments, the sensor **100** may include a drug-eluting polymer matrix that disperses one or more therapeutic agents (e.g., an anti-inflammatory drug).

(19) In some embodiments, as illustrated in FIG. 2, the sensor **100** may include a substrate **116**. In some embodiments, the substrate **116** may be a circuit board (e.g., a printed circuit board (PCB) or flexible PCB) on which circuit components (e.g., analog and/or digital circuit components) may be mounted or otherwise attached. However, in some alternative embodiments, the substrate **116** may be a semiconductor substrate having circuitry fabricated therein. The circuitry may include analog and/or digital circuitry. Also, in some semiconductor substrate embodiments, in addition to the circuitry fabricated in the semiconductor substrate, circuitry may be mounted or otherwise attached to the semiconductor substrate **116**. In other words, in some semiconductor substrate embodiments, a portion or all of the circuitry, which may include discrete circuit elements, an integrated circuit (e.g., an application specific integrated circuit (ASIC)) and/or other electronic components (e.g., a non-volatile memory), may be fabricated in the semiconductor substrate **116** with the remainder of the circuitry is secured to the semiconductor substrate **116** and/or a core (e.g., ferrite core) for the inductor **114**. In some embodiments, the semiconductor substrate **116** and/or a core may provide communication paths between the various secured components.

(20) In some embodiments, the one or more of the sensor housing **102**, analyte indicator **106**, indicator molecules **104**, light source **108**, photodetectors **224**, **226**, temperature transducer **670**, substrate **116**, and inductor **114** of sensor **100** may include some or all of the features described in one or more of U.S. application Ser. No. 13/761,839, filed on Feb. 7, 2013, U.S. application Ser. No. 13/937,871, filed on Jul. 9, 2013, and U.S. application Ser. No. 13/650,016, filed on Oct. 11, 2012, all of which are incorporated by reference in their entireties. Similarly, the structure and/or function of the sensor **100** and/or transceiver **101** may be as described in one or more of U.S. application Ser. Nos. 13/761,839, 13/937,871, and 13/650,016.

(21) Although in some embodiments, as illustrated in FIG. 2, the sensor **100** may be an optical sensor, this is not required, and, in one or more alternative embodiments, sensor **100** may be a different type of analyte sensor, such as, for example, an electrochemical sensor, a diffusion sensor, or a pressure sensor. Also, although in some embodiments, as illustrated in FIGS. 1 and 2, the analyte sensor **100** may be a fully implantable sensor, this is not required, and, in some alternative embodiments, the sensor **100** may be a transcutaneous sensor having a wired connection to the transceiver **101**. For example, in some alternative embodiments, the sensor **100** may be located in or on a transcutaneous needle (e.g., at the tip thereof). In these embodiments, instead of wirelessly communicating using inductors **103** and **114**, the sensor **100** and transceiver **101** may communicate

using one or more wires connected between the transceiver **101** and the transceiver transcutaneous needle that includes the sensor **100**. For another example, in some alternative embodiments, the sensor **100** may be located in a catheter (e.g., for intravenous blood glucose monitoring) and may communicate (wirelessly or using wires) with the transceiver **101**.

(22) In some embodiments, the sensor **100** may include a transceiver interface device. In some embodiments where the sensor **100** includes an antenna (e.g., inductor **114**), the transceiver interface device may include the antenna (e.g., inductor **114**) of sensor **100**. In some of the transcutaneous embodiments where there exists a wired connection between the sensor **100** and the transceiver **101**, the transceiver interface device may include the wired connection.

(23) FIGS. **3** and **4** are cross-sectional and exploded views, respectively, of a non-limiting embodiment of the transceiver **101**, which may be included in the analyte monitoring system illustrated in FIG. **1**. As illustrated in FIG. **4**, in some non-limiting embodiments, the transceiver **101** may include a graphic overlay **204**, front housing **206**, button **208**, printed circuit board (PCB) assembly **210**, battery **212**, gaskets **214**, antenna **103**, frame **218**, reflection plate **216**, back housing **220**, ID label **222**, and/or vibration motor **928**. In some non-limiting embodiments, the vibration motor **928** may be attached to the front housing **206** or back housing **220** such that the battery **212** does not dampen the vibration of vibration motor **928**. In a non-limiting embodiment, the transceiver electronics may be assembled using standard surface mount device (SMD) reflow and solder techniques. In one embodiment, the electronics and peripherals may be put into a snap together housing design in which the front housing **206** and back housing **220** may be snapped together. In some embodiments, the full assembly process may be performed at a single external electronics house. However, this is not required, and, in alternative embodiments, the transceiver assembly process may be performed at one or more electronics houses, which may be internal, external, or a combination thereof. In some embodiments, the assembled transceiver **101** may be programmed and functionally tested. In some embodiments, assembled transceivers **101** may be packaged into their final shipping containers and be ready for sale.

(24) In some embodiments, as illustrated in FIGS. **3** and **4**, the antenna **103** may be contained within the housing **206** and **220** of the transceiver **101**. In some embodiments, the antenna **103** in the transceiver **101** may be small and/or flat so that the antenna **103** fits within the housing **206** and **220** of a small, lightweight transceiver **101**. In some embodiments, the antenna **103** may be robust and capable of resisting various impacts. In some embodiments, the transceiver **101** may be suitable for placement, for example, on an abdomen area, upper-arm, wrist, or thigh of a patient body. In some non-limiting embodiments, the transceiver **101** may be suitable for attachment to a patient body by means of a biocompatible patch. Although, in some embodiments, the antenna **103** may be contained within the housing **206** and **220** of the transceiver **101**, this is not required, and, in some alternative embodiments, a portion or all of the antenna **103** may be located external to the transceiver housing. For example, in some alternative embodiments, antenna **103** may wrap around a user's wrist, arm, leg, or waist such as, for example, the antenna described in U.S. Pat. No. 8,073,548, which is incorporated herein by reference in its entirety.

(25) FIG. **5** is a schematic view of an external transceiver **101** according to a non-limiting embodiment. In some embodiments, the transceiver **101** may have a connector **902**, such as, for example, a Micro-Universal Serial Bus (USB) connector. The connector **902** may enable a wired connection to an external device, such as a personal computer (e.g., personal computer **109**) or a display device **105** (e.g., a smartphone).

(26) The transceiver **101** may exchange data to and from the external device through the connector **902** and/or may receive power through the connector **902**. The transceiver **101** may include a connector integrated circuit (IC) **904**, such as, for example, a USB-IC, which may control transmission and receipt of data through the connector **902**. The transceiver **101** may also include a charger IC **906**, which may receive power via the connector **902** and charge a battery **908** (e.g., lithium-polymer battery). In some embodiments, the battery **908** may be rechargeable, may have a

short recharge duration, and/or may have a small size.

(27) In some embodiments, the transceiver **101** may include one or more connectors in addition to (or as an alternative to) Micro-USB connector **904**. For example, in one alternative embodiment, the transceiver **101** may include a spring-based connector (e.g., Pogo pin connector) in addition to (or as an alternative to) Micro-USB connector **904**, and the transceiver **101** may use a connection established via the spring-based connector for wired communication to a personal computer (e.g., personal computer **109**) or a display device **105** (e.g., a smartphone) and/or to receive power, which may be used, for example, to charge the battery **908**.

(28) In some embodiments, the transceiver **101** may have a wireless communication IC **910**, which enables wireless communication with an external device, such as, for example, one or more personal computers (e.g., personal computer **109**) or one or more display devices **105** (e.g., a smartphone). In one non-limiting embodiment, the wireless communication IC **910** may employ one or more wireless communication standards to wirelessly transmit data. The wireless communication standard employed may be any suitable wireless communication standard, such as an ANT standard, a Bluetooth™ standard, or a Bluetooth™ Low Energy (BLE) standard (e.g., BLE 4.0). In some non-limiting embodiments, the wireless communication IC **910** may be configured to wirelessly transmit data at a frequency greater than 1 gigahertz (e.g., 2.4 or 5 GHz). In some embodiments, the wireless communication IC **910** may include an antenna (e.g., a Bluetooth™ antenna). In some non-limiting embodiments, the antenna of the wireless communication IC **910** may be entirely contained within the housing (e.g., housing **206** and **220**) of the transceiver **101**. However, this is not required, and, in alternative embodiments, all or a portion of the antenna of the wireless communication IC **910** may be external to the transceiver housing.

(29) In some embodiments, the transceiver **101** may include a display interface device, which may enable communication by the transceiver **101** with one or more display devices **105**. In some embodiments, the display interface device may include the antenna of the wireless communication IC **910** and/or the connector **902**. In some non-limiting embodiments, the display interface device may additionally include the wireless communication IC **910** and/or the connector IC **904**.

(30) In some embodiments, the transceiver **101** may include voltage regulators **912** and/or a voltage booster **914**. The battery **908** may supply power (via voltage booster **914**) to a radio-frequency identification (RFID) reader IC **916**, which may use an inductor **103** to convey information (e.g., commands) to the sensor **101** and receive information (e.g., measurement information) from the sensor **100**. In some non-limiting embodiments, the sensor **100** and transceiver **101** may communicate using near field communication (NFC) (e.g., at a frequency of 13.56 MHz). In some embodiments, as shown in FIG. 5, the inductor **103** may be a flat antenna **919**. In some non-limiting embodiments, the inductor **103** may be flexible. However, as noted above, the inductor **103** of the transceiver **101** may be in any configuration that permits adequate field strength to be achieved when brought within adequate physical proximity to the inductor **114** of the sensor **100**. In some embodiments, the transceiver **101** may include a power amplifier **918** to amplify the signal to be conveyed by the inductor **103** to the sensor **100**.

(31) In some embodiments, the transceiver **101** may include a peripheral interface controller (PIC) microcontroller **920** and memory **922** (e.g., Flash memory), which may be non-volatile and/or capable of being electronically erased and/or rewritten. The PIC microcontroller **920** may control the overall operation of the transceiver **101**. For example, the PIC microcontroller **920** may control the connector IC **904** or wireless communication IC **910** to transmit data via wired or wireless communication and/or control the RFID reader IC **916** to convey data via the inductor **103**. The PIC microcontroller **920** may also control processing of data received via the inductor **103**, connector **902**, or wireless communication IC **910**.

(32) In some embodiments, the transceiver **101** may include a sensor interface device, which may enable communication by the transceiver **101** with a sensor **100**. In some embodiments, the sensor interface device may include the inductor **103**. In some non-limiting embodiments, the sensor

interface device may additionally include the RFID reader IC **916** and/or the power amplifier **918**. However, in some alternative embodiments where there exists a wired connection between the sensor **100** and the transceiver **101** (e.g., transcutaneous embodiments), the sensor interface device may include the wired connection.

(33) In some embodiments, the transceiver **101** may include a user interface including one or more of a display **924**, a speaker **926**, and a vibration motor **928**. In some embodiments, the display device **924** may be a liquid crystal display and/or one or more light emitting diodes, and the PIC microcontroller **920** may control the display **924** to display data (e.g., analyte concentration values). In some embodiments, the speaker **926** (e.g., a beeper) and/or the vibration motor **928** may be activated, for example, in the event that an alarm condition (e.g., detection of a hypoglycemic or hyperglycemic condition) is met. The transceiver **101** may also include one or more additional sensors **930**, which may include an accelerometer and/or temperature sensor, that may be used in the processing performed by the PIC microcontroller **920**.

(34) In some embodiments, the transceiver **101** may be a body-worn transceiver that is a rechargeable, external device worn over the sensor implantation or insertion site. The transceiver **101** may supply power to the proximate sensor **100**, calculate analyte concentrations from data received from the sensor **100**, and/or transmit the calculated analyte concentrations to a display device **105** (see FIG. 1). Power may be supplied to the sensor **100** through an inductive link (e.g., an inductive link of 13.56 MHz). In some embodiments, the transceiver **101** may be placed using an adhesive patch or a specially designed strap or belt. The external transceiver **101** may read measured analyte data from a subcutaneous sensor **100** (e.g., up to a depth of 2 cm or more). The transceiver **101** may periodically (e.g., every 2, 5, or 10 minutes) read sensor data and calculate an analyte concentration and an analyte concentration trend. From this information, the transceiver **101** may also determine if an alert and/or alarm condition exists, which may be signaled to the user (e.g., through vibration by vibration motor **928** and/or an LED of the transceiver's display **924** and/or a display of a display device **105**). The information from the transceiver **101** (e.g., calculated analyte concentrations, calculated analyte concentration trends, alerts, alarms, and/or notifications) may be transmitted to a display device **105** (e.g., via Bluetooth Low Energy with Advanced Encryption Standard (AES)-Counter CBC-MAC (CCM) encryption) for display by an application being executed by the display device **105**. In other embodiments, the information from the transceiver **101** may be transmitted to a personal computer (PC) **109** or other secondary display devices (not shown) connected over a network.

(35) In some embodiments, the transceiver **101** of the analyte monitoring system **50** may receive raw signals indicative of an amount or concentration of an analyte in proximity to the analyte indicator **106** of the analyte sensor **100**. In some embodiments, the transceiver **101** may receive the raw signals from the sensor **100** periodically (e.g., every 5, 10, or 20 minutes). In some embodiments, the raw signals may include one or more measurements (e.g., one or more measurements indicative of the level of emission light **331** from the indicator molecules **104** as measured by the photodetector **224**, one or more measurements indicative of the level of reference light **333** as measured by photodetector **226**, and/or one or more temperature measurements as measured by the temperature transducer **670**). In some embodiments, the transceiver **101** may use the received raw signals to calculate analyte concentration. In some embodiments, the transceiver **100** may store one or more calculated analyte concentrations (e.g., in memory **922**). In some embodiments, the transceiver **100** may convey one or more calculated analyte concentrations to the display device **105**, and the display device **105** may display the one or more calculated analyte concentrations.

(36) In some embodiments, the analyte monitoring system **50** may calibrate the conversion of raw signals to analyte concentration. In some embodiments, the calibration may be performed approximately periodically (e.g., approximately every 12 or 24 hours). In some embodiments, the calibration may be performed using one or more reference measurements (e.g., one or more self-

monitoring blood glucose (SMBG) measurements), which may be entered into the analyte monitoring system **50** using the user interface of the display device **105**. In some embodiments, the transceiver **101** may receive the one or more reference measurements from the display device **105** and perform the calibration. One or more of the reference measurements may be erroneous and may lead to erroneous analyte measurement calculation if used as a calibration point for the calibrating of the conversion of raw sensor data to analyte measurements. Accordingly, the analyte monitoring system **5** (e.g., the transceiver **101**) may determine whether to accept (or reject) reference measurements as calibration points in the calibration process. This calibration point acceptance process may be used to prevent erroneous reference measurements from being used as calibration points when calibrating the function used to convert raw sensor data (e.g., light and/or temperature measurements) into analyte measurements (e.g., analyte concentrations). In this way, the calibration point acceptance process may increase the accuracy and/or precision of the analyte measurements.

(37) In some embodiments, the transceiver **101** may store the measurement information received from the sensor **100** (e.g., in memory **922**). As noted above, the measurement information received from the sensor **100** may include one or more of: (i) a signal channel measurement with light source **108** on, (ii) a reference or second signal channel measurement with light source **108** on, (iii) a light source current source voltage measurement, (iv) field current measurement, (v) a diagnostic measurement, (vi) an ambient signal channel measurement with light source **108** off, (vii) an ambient reference or second signal channel measurement with light source **108** off, and (viii) a temperature measurement.

(38) In some embodiments, the transceiver **101** may additionally store (e.g., in memory **922**) other data with the measurement information received from the sensor **100**. In some non-limiting embodiments, the other data may include one or more of: (i) an analyte concentration (e.g., in mg/dL, such as, for example, within a range of 20.0 to 400.0 mg/dL) calculated by the transceiver **101** from the measurement information, (ii) the date and time that the analyte measurement was taken, (iii) accelerometer values (e.g., x, y, and z) taken from an accelerometer of the transceiver **101** (e.g., an accelerometer of additional sensors **930**), and/or (iv) the temperature of the transceiver **101** as measured by a temperature sensor of the transceiver **101** (e.g., a temperature sensor of additional sensors **930**). In some embodiments, the transceiver **101** may keep track of the date and time and, as noted above, store the date and time along with the received analyte measurement information and/or calculated analyte concentration. In embodiments where the transceiver **101** includes an accelerometer, the accelerometer will enable tracking of activity levels of the subject that is wearing the transceiver **101**. This activity level may be included in an event log and incorporated into various algorithms (e.g., for analyte concentration calculation, trending, and/or contributing to potential dosing levels for the subjects). In some embodiments, the transceiver **101** may store (e.g., in memory **922**) any alert and/or alarm conditions detected based on the calculated analyte concentrations.

(39) In some embodiments, the transceiver **101** may have a power button (e.g., button **208**) to allow the user to turn the device on or off, reset the device, or check the remaining battery life. In some embodiments, the transceiver **101** may have a button, which may be the same button as a power button or an additional button, to suppress one or more user notification signals (e.g., vibration, visual, and/or audible) of the transceiver **101** generated by the transceiver **101** in response to detection of an alert or alarm condition.

(40) FIG. **6** is a block diagram of a non-limiting embodiment of the display device **105** of the analyte monitoring system **50**. As shown in FIG. **6**, in some embodiments, the display device **105** may include one or more of a connector **302**, a connector integrated circuit (IC) **304**, a charger IC **306**, a battery **308**, a computer **310**, a first wireless communication IC **312**, a memory **314**, a second wireless communication IC **316**, a third wireless communication IC **318**, and a user interface **340**.

(41) In some embodiments in which the display device **105** includes the connector **302**, the connector **302** may be, for example and without limitation, a Micro-Universal Serial Bus (USB) connector. The connector **302** may enable a wired connection to an external device, such as a personal computer or transceiver **101**. The display device **105** may exchange data to and from the external device through the connector **302** and/or may receive power through the connector **302**. In some embodiments, the connector IC **304** may be, for example and without limitation, a USB-IC, which may control transmission and receipt of data through the connector **302**.

(42) In some embodiments in which the display device **105** includes the charger IC **306**, the charger IC **306** may receive power via the connector **302** and charge the battery **308**. In some non-limiting embodiments, the battery **308** may be, for example and without limitation, a lithium-polymer battery. In some embodiments, the battery **308** may be rechargeable, may have a short recharge duration, and/or may have a small size.

(43) In some embodiments, the display device **105** may include one or more connectors and/or one or more connector ICs in addition to (or as an alternative to) connector **302** and connector IC **304**. For example, in some alternative embodiments, the display device **105** may include a spring-based connector (e.g., Pogo pin connector) in addition to (or as an alternative to) connector **302**, and the display device **105** may use a connection established via the spring-based connector for wired communication to a personal computer or the transceiver **101** and/or to receive power, which may be used, for example, to charge the battery **308**.

(44) In some embodiments in which the display device **105** includes the first wireless communication IC **312**, the first wireless communication IC **312** may enable wireless communication with one or more external devices, such as, for example, one or more personal computers, one or more transceivers **101**, and/or one or more other display devices **105**. In some non-limiting embodiments, the first wireless communication IC **312** may employ one or more wireless communication standards to wirelessly transmit data. The wireless communication standard employed may be any suitable wireless communication standard, such as an ANT standard, a Bluetooth standard, or a Bluetooth Low Energy (BLE) standard (e.g., BLE 4.0). In some non-limiting embodiments, the first wireless communication IC **312** may be configured to wirelessly transmit data at a frequency greater than 1 gigahertz (e.g., 2.4 or 5 GHz). In some embodiments, the first wireless communication IC **312** may include an antenna (e.g., a Bluetooth antenna). In some non-limiting embodiments, the antenna of the first wireless communication IC **312** may be entirely contained within a housing of the display device **105**. However, this is not required, and, in alternative embodiments, all or a portion of the antenna of the first wireless communication IC **312** may be external to the display device housing.

(45) In some embodiments, the display device **105** may include a transceiver interface device, which may enable communication by the display device **105** with one or more transceivers **101**. In some embodiments, the transceiver interface device may include the antenna of the first wireless communication IC **312** and/or the connector **302**. In some non-limiting embodiments, the transceiver interface device may additionally or alternatively include the first wireless communication IC **312** and/or the connector IC **304**.

(46) In some embodiments in which the display device **105** includes the second wireless communication IC **316**, the second wireless communication IC **216** may enable the display device **105** to communicate with one or more remote devices (e.g., smartphones, servers, and/or personal computers) via wireless local area networks (e.g., Wi-Fi), cellular networks, and/or the Internet. In some non-limiting embodiments, the second wireless communication IC **316** may employ one or more wireless communication standards to wirelessly transmit data. In some embodiments, the second wireless communication IC **316** may include one or more antennas (e.g., a Wi-Fi antenna and/or one or more cellular antennas). In some non-limiting embodiments, the one or more antennas of the second wireless communication IC **316** may be entirely contained within a housing of the display device **105**. However, this is not required, and, in alternative embodiments, all or a

portion of the one or more antennas of the second wireless communication IC **316** may be external to the display device housing.

(47) In some embodiments, in which the display device **105** includes the third wireless communication IC **318**, the third wireless communication IC **318** may enable the display device **105** to communicate directly with the sensor **100** so that the display device **105** may additionally perform some or all of the functions of the transceiver **101**. In some embodiments, the display device **105** and the sensor **100** may communicate using NFC (e.g. at a frequency of 13.56 MHz). In some embodiments, the display device **105** may include an inductor (e.g. flat antenna, loop antenna, etc.) that is configured to permit adequate field strength to be achieved when brought within adequate physical proximity to the inductor **114** of the sensor **100**. In some non-limiting embodiments, the display device **105** may receive sensor data from the sensor **100** periodically (e.g., every 1, 2, 5, 10, 15, or 20 minutes). In some non-limiting embodiments, the display device **105** may receive sensor data from the sensor **100** on demand (e.g., when the display device **100** is hovered or swiped in proximity to the sensor **100**). In some non-limiting embodiments, the display device **105** may include a sensor interface device, which may enable communication by the display device **105** with a sensor **100**. In some embodiments, the sensor interface device may include the inductor. In some non-limiting embodiments, the sensor interface device may additionally include the RFID reader IC **916** and/or the power amplifier **918** described above with reference to FIG. 5

(48) In some embodiments in which the display device **105** includes the memory **314**, the memory **314** may be non-volatile and/or capable of being electronically erased and/or rewritten. In some embodiments, the memory **314** may be, for example and without limitations a Flash memory.

(49) In some embodiments in which the display device **105** includes the computer **310**, the computer **310** may control the overall operation of the display device **105**. For example, the computer **310** may control the connector IC **304**, the first wireless communication IC **312**, the second wireless communication IC **316**, and/or the third wireless communication IC **318** to transmit data via wired or wireless communication. The computer **310** may additionally or alternatively control processing of received data (e.g., analyte monitoring data received from the transceiver **101**).

(50) In some embodiments in which the display device **105** includes the user interface **340**, the user interface **340** may include one or more of a display **320** and a user input **322**. In some embodiments, the display **320** may be a liquid crystal display (LCD) and/or light emitting diode (LED) display. In some non-limiting embodiments, the user input **322** may include one or more buttons, a keyboard, a keypad, and/or a touchscreen. In some embodiments, the computer **310** may control the display **320** to display data (e.g., analyte concentration values, analyte trend information, alerts, alarms, and/or notifications). In some embodiments, the user interface **340** may include one or more of a speaker **324** (e.g., a beeper) and a vibration motor **326**, which may be activated, for example, in the event that a condition (e.g., a hypoglycemic or hyperglycemic condition) is met.

(51) In some embodiments, the computer **310** may execute a mobile medical application (MMA). In some embodiments, the display device **105** may receive analyte monitoring data from the transceiver **101**. The received analyte monitoring data may include one or more analyte concentrations, one or more analyte concentrations trends, and/or one or more sensor measurements. The received analyte monitoring data may additionally or alternatively include alarms, alerts, and/or notifications. In some embodiments, the display device **105** may receive measured analyte data directly from the sensor **100**. The display device **105** may calculate an analyte concentration and an analyte concentration trend using at least the received sensor data. From this analyte information, the display device **105** may also determine if an alert and/or alarm condition exists, which may be signaled to the user (e.g., through vibration by a vibration motor and/or a display of a display device **105**). In some embodiments, this analyte information (e.g., calculated analyte concentrations, calculated analyte concentration trends, alerts, alarms, and/or

notifications) may be displayed by the MMA being executed by the display device **105**. In some embodiments, the display device **105** may transmit this information (e.g., calculated analyte concentrations, calculated analyte concentration trends, alerts, alarms, and/or notifications) over a network such that a remote computing device (e.g., server) and one or more secondary display devices may receive, store, and display the analyte information.

(52) In some embodiments, the analyte monitoring system **50** may calibrate the conversion of raw sensor measurements to analyte concentrations. In some embodiments, the calibration may be performed approximately periodically (e.g., every 12 or 24 hours). In some embodiments, the calibration may be performed using one or more reference measurements (e.g., one or more self-monitoring blood glucose (SMBG) measurements). In some embodiments, the reference measurements may be entered into the analyte monitoring system **50** using the user interface **340** of the display device **105**. In some embodiments, the display device **105** may convey one or more reference measurements to the transceiver **101**, and the transceiver **101** may use the one or more received reference measurements to perform the calibration. In some embodiments, the display device may additionally or alternatively use the one or more reference measurements to perform a calibration.

(53) FIG. 7 is a block diagram of a non-limiting embodiment of the computer **310** of the analyte monitoring system **50**. As shown in FIG. 7, in some embodiments, the computer **310** may include one or more processors **522** (e.g., a general purpose microprocessor) and/or one or more circuits, such as an application specific integrated circuit (ASIC), field-programmable gate arrays (FPGAs), a logic circuit, and the like. In some embodiments, the computer **310** may include a data storage system (DSS) **523**. The DSS **523** may include one or more non-volatile storage devices and/or one or more volatile storage devices (e.g., random access memory (RAM)). In embodiments where the computer **310** includes a processor **522**, the DSS **523** may include a computer program product (CPP) **524**. CPP **524** may include or be a computer readable medium (CRM) **526**. The CRM **526** may store a computer program (CP) **528** comprising computer readable instructions (CRI) **530**. In some embodiments, the CRM **526** may store, among other programs, the MMA, and the CRI **530** may include one or more instructions of the MMA. The CRM **526** may be a non-transitory computer readable medium, such as, but not limited, to magnetic media (e.g., a hard disk), optical media (e.g., a DVD), solid state devices (e.g., random access memory (RAM) or flash memory), and the like. In some embodiments, the CRI **530** of computer program **528** may be configured such that when executed by processor **522**, the CRI **530** causes the computer **310** to perform steps described below (e.g., steps described below with reference to the MMA). In other embodiments, the computer **310** may be configured to perform steps described herein without the need for a computer program. That is, for example, the computer **310** may consist merely of one or more ASICs. Hence, the features of the embodiments described herein may be implemented in hardware and/or software.

(54) In some embodiments in which the user interface **340** of the display device **105** includes the display **318**, the MMA may cause the display device **105** to provide a series of graphical control elements or widgets in the user interface **340**, such as a graphical user interface (GUI), shown on the display **318**. The MMA may, for example without limitation, cause the display device **105** to display analyte related information in a GUI such as, but not limited to: one or more of analyte information, current analyte concentrations, past analyte concentrations, predicted analyte concentrations, user notifications, analyte status alerts and alarms, trend graphs, arrows, and user-entered events. In some embodiments, the MMA may provide one or more graphical control elements that may allow a user to manipulate aspects of the one or more display screens. Although aspects of the MMA are illustrated and described in the context of glucose monitoring system embodiments, this is not required, and, in some alternative embodiments, the MMA may be employed in other types of analyte monitoring systems.

(55) In some embodiments where the display device **105** communicates with a transceiver **101**,

which in turn obtains sensor measurement data from the analyte sensor **100**, the MMA may cause the display device **105** to receive and display one or more of glucose data, trends, graphs, alarms, and alerts from the transceiver **101**. In some embodiments where the display device **105** communicates directly with the sensor **100** to obtain sensor measurement data, the MMA may cause the display device **105** to receive and display one or more of glucose data, trends, graphs, alarms, and alerts from the transceiver **101**. In some embodiments, the MMA may store glucose level history and statistics for a patient on the display device **105** (e.g., in memory **314** and/or DSS **533**) and/or in a remote data storage system.

(56) In some embodiments, a user of the display device **105**, which may be the same or different individual as patient, may initiate the download of the MMA from a central repository over a wireless cellular network or packet-switched network, such as the Internet. Different versions of the MMA may be provided to work with different commercial operating systems, such as the Android OS or Apple OS running on commercial smart phones, tablets, and the like. For example, where display device **105** is an Apple iPhone, the user may cause the display device **105** to access the Apple iTunes store to download a MMA compatible with the Apple OS, whereas where the display device **105** is an Android mobile device, the user may cause the display device **105** to access the Android App Store to download a MMA compatible with the Android OS.

(57) FIG. **8** is an example of a home screen display of a medical mobile application (MMA) in accordance with aspects of various embodiments of the present invention. According to some embodiments, the workspace display of the MMA may be depicted in a GUI on the display **320** of the display device **105**. In some embodiments, the home screen may display one or more of real-time analyte concentrations either received from transceiver **101** or calculated by the display device **105**, rate and direction of analyte level change, graphical trends of analyte levels, alarms or alerts for hypoglycemia or hyperglycemia, and logged events such as, for example and without limitation, meals, exercise, and medications. Table 1 below depicts several informational non-limiting examples of items and features that may be depicted on the home screen.

(58) TABLE-US-00001

TABLE 1	Home Screen
Status bar	Shows the status of user's glucose level
Transceiver/Transmitter	This is the transceiver being used; the transceiver name ID can be changed by going to Settings > System
Current glucose value	A real-time glucose reading; this may be updated every 5 minutes
Date and time	The current date and time with navigational options, such as scroll left or right to see different dates and times
Alarm and Events	Shows an icon when an alert, alarm, or event occurs
Bluetooth Connection	Shows the strength of the Bluetooth connection
Handheld Device Battery	Indicates the battery strength of the handheld device
Level Transmitter/Transceiver	Indicates the battery strength of the transceiver
Battery Level Transmitter/Transceiver	Shows the strength of the transceiver connection
Connection Status Icon	Trend Arrow
Shows the direction a patient's glucose level is trending	Unit of Measurement
This is the units for the glucose value	High Glucose Alarm
This is the high glucose alarm or alert level set by a user	Level Glucose High Target
This is the high glucose target level set by a user	Level Stacked Alerts
Shows when there are several alerts at the same time	Glucose Trend Graph
A user can navigate or scroll through the graph to see the trend over time	Menu
Navigation to various sections of the MMA, such as: Home Reports Settings Calibrate Share My Data About Notifications Placement Guide Event Log Connect Calibration Point Icon	This icon appears when a calibration is entered
Profile Indicator	This indicator may indicate what profile is being applied, such as a normal profile, temporary profile, vacation profile, and the like.

(59) In some embodiments, as shown in FIG. **8**, the home screen may include one or more of a status notification bar **1301**, a real-time current glucose level **1303** of a patient, one or more icons **1305**, a trend arrow **1307**, a historical graph **1309**, a profile indicator **1333**, and navigation tools **1311**. The status notification bar **1301** may depict, for example and without limitation, alarms, alerts, and notifications related to, for example, glucose levels and system statistics and/or status. The one or more icons **1305** may represent the signal strength of the transceiver **101** and/or the

battery level of the transceiver **101**. The trend arrow **1307** may indicate a rate and/or direction of change in glucose measurements of a patient. The historical graph may be, for example and without limitation, a line graph and may indicate trends of glucose measurement levels of a patient. The navigation tools **1311** may allow a user to navigate through different areas or screens of the MMA. The screens may include, for example and without limitation, one or more of Home, Calibrate, Event Log, Notifications, and Menu screens.

(60) In some embodiments, the historical graph **1309** may depict logged events and/or user inputted activities such as meals (nutrition, amount of carbohydrates), exercise (amount of exercise), medication (amount of insulin units), and blood glucose values as icons on positions of the graph corresponding to when such events occurred. In some embodiments, the historical graph **1309** may show one or more of a boundary or indication of a high glucose alarm level **1313**, a low glucose alarm level **1315**, a high glucose target level **1317**, and a low glucose target level **1319**. In some embodiments, a user may interact with a time or date range **1321** option via the GUI to adjust the time period of the glucose level displayed on the historical graph **1309**. In some embodiments, the date range **1321** may be specified by a user and may be set to different time periods such as 1, 3, 24 hours, 1, 7, 14, 30, and 60 days, weeks, months, etc. In some embodiments, the line graph **1309** may show high, low, and average glucose levels of a patient for the selected date range **1321**. In other embodiments, the line graph **1309** may be a pie chart, log book, modal day, or other depiction of glucose levels of a patient over a selectable date range **1321**, any of which may further depict high, low, and average glucose levels of the patient over that date range **1321**.

(61) In some non-limiting embodiments, the trend arrow **1307** may be depicted in five different configurations that signify direction (up, down, neutral) and rate (rapidly, very rapidly, slow, very slow, and stable) of glucose change. In some non-limiting embodiments, the MMA and/or the transceiver **101** may use the last twenty minutes of continuous glucose measurement data received from the sensor **101** and/or processed by the transceiver **730** in the calculation used to determine the orientation of the trend arrow **1307**. In some embodiments, there may be times when the trend arrow **1307** may not be displayed due to, for example, there being insufficient sensor values available for the trend calculation. In some embodiments, a trend arrow **1307** displayed in a horizontal orientation (approximately 0° along the horizontal direction of the GUI display) may indicate that the glucose level is changing gradually, such as, for example, at a rate between -1.0 mg/dL and 1.0 mg/dL per minute. In some embodiments, a trend arrow **1307** displayed slightly in the upwards direction (approximately 45° up from the horizontal direction of the GUI display) may indicate that the glucose level is rising moderately, such as, for example, at a rate between 1.0 mg/dL and 2.0 mg/dL per minute. In some embodiments, a trend arrow **1307** displayed slightly in the downwards direction (approximately 45° down from the horizontal direction of the GUI display) may indicate that the glucose level is falling moderately, such as, for example, at a rate between 1.0 mg/dL and 2.0 mg/dL per minute. In some embodiments, a trend arrow **1307** displayed in a vertical direction (approximately 90° up from the horizontal direction of the GUI display) may indicate that the glucose level is rising very rapidly, such as, for example, at a rate more than 2.0 mg/dL per minute. In some embodiments, a trend arrow **1307** displayed in a downwards direction (approximately 90° down from the horizontal direction of the GUI display) may indicate that the glucose level is falling very rapidly, such as, for example, at a rate more than 2.0 mg/dL per minute. In some embodiments, the trend arrow **1307** is different from a predicted glucose alarm or alert. For example, the trend arrow **1307** may indicate rate and direction of change regardless of glucose value, whereas predicted glucose alarms or alerts may indicate reaching a certain glucose level based on current trends. For example, the MMA may cause a predicted low glucose alarm or alert to be displayed in the notification bar **1301** while still displaying a relatively stable trend arrow **1307** (e.g., at 0° or 45° from the horizontal direction of the GUI display).

(62) In some embodiments, the historical line graph **1309** may allow user to quickly review and analyze historical data and/or trend information of a patient's glucose levels over time. In some

embodiments, the historical line graph **1309** may include icons or markers along the trend line to reflect alarms, alerts, notifications, and/or any events that were automatically or manually logged by the user into the display device **105** via a GUI display generated by the MMA. Where one or more of such icons or markers are displayed on the historical line graph **1309**, a user may select any one of the icons or markers to obtain more information about the item. For example, in response to a selection of a mark on the line graph **1309**, the MMA may generate a popup window on the display **220** that provides more information about the mark.

(63) In some embodiments, the historical line graph **1309** may enable a user to quickly review how well a patient is doing against glucose targets and/or alarms or alerts. For example, a user may establish a high glucose alarm level **1313** and/or a low glucose alarm level **1315**, as well as a high glucose target level **1317** and/or a low glucose target level **1319**. The high glucose alarm level **1313** and/or low glucose alarm level **1315** may be visually depicted over the historical line graph **1309**, for example, using a colored dashed line (such as red). Additionally, the high glucose target level **1317** and low glucose target level **1319** may be visually depicted over the historical line graph **1309**, for example, using a color dashed line (such as green).

(64) In some embodiments, the colors of the historical line graph **1309** may change depending on a glucose level **1303** status. For example, during the times where the glucose level **1303** was outside of the high glucose alarm level **1313** or low glucose alarm level **1315**, then the portion of the line graph **1309** corresponding to those times may be filled in red. As another example, during the times where the glucose level **1303** is between the high glucose target level **1317** and the low glucose target level **1319**, then the portion of the line graph **1309** corresponding to those times may be filled in green. As yet another example, during the times where the glucose level **1303** is between a glucose target level **1317**, **1319** and a corresponding alarm level **1313**, **1315**, then the portion of the line graph **1309** may be filled in yellow.

(65) In some embodiments, the line graph **1309** may be displayed with one or more selectable date range icons **1321** that allow a user to change the day/time period corresponding to the line graph **1309** in real-time. For example, a user may select a forwards or backwards selectable option (such as an arrow) or use a swipe or fling gesture that may be recognized by GUI to navigate to a later or earlier time period, respectively, such as a day, month, etc. In some embodiments a user may choose an older graph **1309** to display by tapping the date on the date range **1321** portion of the screen and submitting or entering a desired date and/or time to review. In some embodiments, a user may use one or more gestures that are recognized by the GUI, such as a pinch, zoom, tap, press and hold, or swipe, on graph **1309**. For example, a user may pinch the historical line graph **1309** with a thumb and index finger in order to cause the MMA to display different time/dating settings or adjust a time/date setting on the line graph **1309**. In some embodiments, a user may tap or press and hold a time event on historical line graph **1309**, and in response the MMA may display further detail on the time event, such as a history, reading value, date/time, or association to other events or display a prompt for entry of a time event.

(66) In some embodiments, the MMA may store glucose data **1303** on the display device **105** (e.g., in memory **214** and/or DSS **533**) so long as there is available memory space. Additionally or alternatively, the MMA may cause the display device **105** to send a sync request message to store the glucose data **1303** on a remote storage device.

(67) In some embodiments, the MMA may cause the GUI to display navigational tools **1311** that allow a user to navigate to different features and screens provided by the MMA. For example, the navigational tools **1311** may include a navigation bar with one or more of a plurality of selectable navigation options **1323**, **1325**, **1327**, **1329**, and **1331**, such as buttons or icons. As shown in FIG. **4**, in some embodiments, the selectable navigation options may allow a user to navigate to one or more of the “Home” screen **1323**, a “Calibrate” screen **1325**, an “Event Log” screen **1327**, a “Notifications” screen **1329**, and a “Menu” screen **1331**. Upon a user selection of one of the selectable navigation options in the navigation tools area **1311**, a new screen corresponding to the

selected option may be displayed on a display device by the GU

(68) In some embodiments where the system includes the data management system (DMS) **111** (see FIG. 1), the DMS **111** may be a web-based analyte DMS. In some embodiments, the DMS **111** may be a server device employed to allow data to be shared over the network such as the Internet. The server may share data via proprietary formats configured to be employed by hardware computing systems configured, at least in part, with applications to make the hardware computing system into an analyte monitoring system. In some embodiments, data from the display device **105** and/or PC **109** may be uploaded (e.g., through a wired connection such as, for example, a USB connection or a wireless connection such as, for example, a wireless Internet connection) to a web server on a remote computer. In some embodiments, the DMS **111** may enable sharing of the analyte data (e.g., allowing the user, caregiver, and/or clinician to view sensor analyte data). The user may collect analyte data at home or in a clinic/research facility and then upload the data to their computer web account. Using the web account, the DMS **111** may use the data to generate one or more different reports utilizing the uploaded information. For example, in some non-limiting embodiments, the DMS **111** may use the uploaded data to generate one or more of the following reports: (i) an analyte details report, (ii) an analyte line report, (iii) a modal day report, (iv) a modal summary report, (v) a statistics report, and (vi) a transceiver log report.

(69) In some embodiments, a user may use the DMS **111** to register with the DMS **111** and create a unique user ID and password. Once logged in, the user may enter their basic user information and may upload analyte reading data from their transceiver **101** or display device **105**. In various embodiments, the DMS **111** may support specific data types such as, for example, glucose, insulin, meal/carbs, exercise, health event, alarms, and errors. In some non-limiting embodiments, data can be automatically uploaded or entered manually by the user or imported from the transceiver **101** and then saved in the DMS **111** to be viewed at a later date.

(70) In some embodiments, the analyte monitor system **50** may be used according to two or more modes including (i) a continuous glucose monitoring (CGM) mode and (ii) a flash glucose monitoring (FGM) mode. In some embodiments, a host or user of the analyte sensor **100** may switch back and forth between using the analyte monitor system **50** according to the the CGM mode and using the analyte monitor system **50** in the FGM mode. In some embodiments, the host or user may use the analyte monitoring system **50** according to the CGM mode by placing the transceiver **101** in proximity to the analyte sensor **100** (e.g., using adhesive or an armband) so that the transceiver **101** and analyte sensor can communicate. In some embodiments, the host or user may use the analyte monitoring system **50** according to the FGM mode by using the display device **105** to communicate directly with the analyte sensor **100**. In some embodiments, when a host or user is using the analyte monitoring system **50** according to the CGM mode, the transceiver **101** may cause the analyte sensor **100** to take one or more measurements and convey the sensor data directly to the transceiver **101**. In some non-limiting embodiments, when the analyte monitoring system **50** is being used according to the CGM mode, the transceiver **101** may cause the analyte sensor **100** to convey sensor data on a periodic basis (e.g., every five minutes, ten minutes, or fifteen minutes). In some embodiments, when a host or user is using the analyte monitoring system **50** according the FGM mode, the display device **105** may cause the analyte sensor **100** to take one or more measurements and convey the sensor data directly to the display device **105**. In some non-limiting embodiments, when the analyte monitoring system **50** is being used according to the FGM mode, the display device **105** cause the analyte sensor **100** to convey sensor data on demand (e.g., by hovering or swiping the display device **105** in proximity to the analyte sensor **100**).

(71) In some embodiments, while the analyte monitor system **50** is being used according to the CGM mode, the transceiver **101** may be operatively linked to the sensor **100**. In some embodiments, under the CGM mode, the transceiver **101** may be placed within adequate physical proximity to the sensor **100** using an adhesive patch, strap, or belt, such that the transceiver **101** is worn on a body of a host. In some embodiments, under the CGM mode, the transceiver **101** may

supply power to the sensor **100** and receive sensor measurement data (e.g., raw signals indicative of an amount or concentration of glucose) from the sensor **100**. In some embodiments, under the CGM mode, the transceiver **101** may periodically use the received sensor measurement data to calculate analyte information, such as for example, one or more analyte concentrations and an analyte concentration trend. In some embodiments, under the CGM mode, the transceiver **101** may also determine if an alert and/or alarm condition exists based on the calculated analyte concentrations and analyte concentration trends. In some embodiments, the transceiver **101** may display the analyte concentrations, the analyte concentration trend, the alert, or the alarm on the display **924** of the transceiver **101**. In some embodiments, the transceiver **101** may notify the user or host about the alert and/or alarm by generating sound with the speaker **926** of the transceiver and/or generating vibration with the vibration motor **928** of the transceiver.

(72) In some embodiments, under the CGM mode, the display device **105** may use the first wireless communication IC **312** to communicate with the transceiver **101** (e.g., via Bluetooth™ Low Energy with Advanced Encryption Standard (AES)-Counter CBC-MAC (CCM) encryption). In some embodiments, the display device **105** may use the first wireless communication IC **312** to receive information from the transceiver **101** (e.g., calculated analyte concentrations, calculated analyte concentration trends, alerts, alarms, and/or notifications). In some embodiments, under the CGM mode, the display device **105** may additionally receive sensor data, which was received by the transceiver **101** from the analyte sensor **100**. In some non-limiting embodiments, under the CGM mode, the transceiver **101** may calculate analyte information using at least the sensor data received directly from the analyte sensor **100**, and the display device **105** may also calculate analyte information using the sensor data received indirectly from the analyte sensor **100** via the transceiver **101**. However, this is not required, and, in some embodiments, only the transceiver **101** calculates analyte information using the sensor data.

(73) In some alternative embodiments, under the CGM mode, only the display device **105** calculates analyte information using sensor data received indirectly from the analyte sensor **100** via the transceiver **101**. In some alternative embodiments, under the CGM mode, the transceiver **101** may convey sensor data received from the analyte sensor **100** to the display device **105** without calculating any analyte information using the sensor data. That is, in some embodiments, under the CGM mode, only the display device **105** (and not the transceiver **101**) calculates analyte information using the sensor data received from the transceiver **101**.

(74) In some embodiments, while the analyte monitor system **50** is being used according to the FGM mode, the display device **105** may be used to communicate directly with the sensor **100**. In some embodiments, under the FGM mode, the display device **105** may use the third wireless communication IC **318** to communicate directly with the sensor (e.g., via NFC at a frequency of 13.56 MHz) and cause the analyte sensor **100** to convey sensor measurement data (e.g., raw signals indicative of an amount or concentration of glucose) to the display device **105**. In some embodiments, under the FGM mode, the display device **105** may cause the analyte sensor **100** to convey sensor measurement data on-demand by positioning (e.g., hovering or swiping) the display device **105** within physical proximity to the sensor **100**. In some embodiments, the positioning the display device **105** in proximity to the analyte sensor **100** may provide adequate coupling between the inductor **114** of the sensor **100** and an inductor of the display device **105**. In some embodiments, the display device **105** may supply power to the sensor **100** when the display device **105** is brought in proximity to the sensor **100**. However, it is not required that the display device **105** supply power to the sensor **100**, and, in some alternative embodiments, the display device **105** may not do so.

(75) In some embodiments, the analyte sensor **100** may be a passive device configured to take measurements only when the transceiver **101** or display device **105** triggers sensor measurements. In some alternative embodiments, the analyte sensor **100** may include a charge storage device (not shown), such as, for example and without limitation, a battery, supercapacitor, or ultracapacitor,

that stores a sufficient amount of energy to maintain the sensor **100** in an operating state even when not operatively linked to either of the transceiver **101** and the display device **105**. The charge storage device may allow the analyte sensor to obtain measurements autonomously (e.g., on a periodic basis). Accordingly, in some embodiments, when the transceiver **101** or display device **101** causes the analyte sensor **100** to convey sensor measurements, the analyte sensor **100** may convey one or more current measurements and/or one or more previous measurements taken by the sensor **100** autonomously.

(76) In some embodiments, under the FGM mode, the display device **105** may use sensor measurement data received directly from the analyte sensor **100** to calculate analyte information, such as for example and without limitation, one or more analyte concentrations and an analyte concentration trend. In some embodiments, the display device **105** may determine whether an alert and/or alarm condition exists based on the calculated analyte concentrations and/or analyte concentration trends.

(77) In some embodiments, while using the analyte monitoring system **50** according to the FGM mode, the transceiver **101** may be operatively disconnected from the sensor **100** such that the transceiver **101** is not within physical proximity to the sensor **100**. In some embodiments, while using the analyte monitoring system **50** according to the FGM mode, the transceiver **101** may not be worn on the body of the host or user of the analyte monitor system **50**. In some embodiments, the display device **105** may be configured to generate a reminder notice to the user or host to cause the analyte sensor **100** to convey sensor data to the display device **105** if the display device **105** has not received sensor data for a predetermined period of time.

(78) In some embodiments, the MMA executed by the display device **105** may cause the display device **105** to display the analyte information (e.g., calculated analyte concentrations, calculated analyte concentration trends, alerts, alarms, and/or notifications). In some embodiments, the display device **105** may display both (i) analyte information generated when the analyte monitoring system **50** is used according to the CGM mode and (ii) analyte information generated when the analyte monitoring system **50** is used according to the FGM mode. In some embodiments, the display device **105** may display (i) analyte information calculated by and received from the transceiver **101** and (ii) analyte information calculated by the display device **105** using at least sensor data received directly from the analyte sensor **100**. In some alternative embodiments, the display device **105** may display (i) analyte information calculated by the display device **105** using at least sensor data received indirectly from the analyte sensor **100** via the transceiver **101** and (ii) analyte information calculated by the display device **105** using at least sensor data received directly from the analyte sensor **100**. In some embodiments, the display device **105** may transmit analyte information (e.g., calculated analyte concentrations, calculated analyte concentration trends, alerts, alarms, and/or notifications) over a network such that a remote computing device (e.g., server) and one or more secondary display devices may receive, store, and display the analyte information.

(79) In some embodiments, while using the analyte monitoring system **50** according to the FGM mode, the transceiver **101** may be operatively disconnected from the sensor **100** such that the transceiver **101** is not within physical proximity to the sensor **100**. In some embodiments, while using the analyte monitoring system **50** according to the FGM mode, the transceiver **101** may not be worn on the body of the host or user of the analyte monitor system **50**. In some embodiments, under the FGM mode, the MMA application is configured to generate a reminder notice to the user or host to cause the analyte sensor **100** to convey sensor data to the display device **105** if the display device **105** has not received sensor data for a predetermined period of time.

(80) FIG. **9** is a flow chart illustrating a process **700** embodying aspects of the present invention. In some embodiments, the process **700** may include one or more steps (e.g., steps **701-703**) in which the analyte monitoring system **50** is used according to the CGM mode and one or more steps (e.g., steps **704-705**) in which the analyte monitoring system **50** is used according to the FGM mode.

(81) In some embodiments, the process **700** may include a step **701** in which a transceiver **101**

receives first sensor data directly from an analyte sensor **100**. In some non-limiting embodiments, in step **701**, the transceiver **101** may cause the sensor to take one or more measurements and convey the first sensor data to the transceiver **101** (e.g., by conveying one or more measurement commands to the analyte sensor **100**). In some embodiments, during step **701**, the transceiver **101** may have been positioned such that the transceiver **101** is operatively linked to the analyte sensor **100**. In some embodiments, the transceiver **101** may have been positioned to be operatively linked to the analyte sensor **100** by wearing the transceiver **101** on an armband, a wrist band, a waist band, or an adhesive patch. In some embodiments, the transceiver **101** may use the inductor **103** and RFID reader IC **916** to receive the second sensor data directly from the analyte sensor **100**. In some embodiments, the transceiver **101** may receive the first sensor data from the analyte sensor **100** wirelessly using a communication standard such as, for example and without limitation, an NFC standard. In some non-limiting embodiments, the analyte sensor **101** may convey the first sensor data by modulating an electromagnetic wave generated by the transceiver **101**, and the transceiver **101** receiving the first sensor data may include detecting the modulations. In some non-limiting embodiments, the analyte sensor **100** may use the electromagnetic wave to power the analyte sensor **100**. In some embodiments, the first sensor data may include one or more raw measurements (e.g., one or more light measurements and one or more temperature measurements).

(82) In some embodiments, the process **700** may include a step **702** in which the transceiver **101** calculates first analyte information using at least the received first sensor data. In some embodiments, the calculated first analyte information may include one or more of an analyte concentration and an analyte concentration trend. In some embodiments, the step **702** of calculating the first analyte information may include determining if an alert and/or alarm condition exists based on at least the calculated first analyte information (e.g., an analyte concentration and/or analyte concentration trend). In some embodiments, the alert and/or alarm conditions may include one or more of a high analyte alarm level, a low analyte alarm level, a high target analyte level, and a low target analyte level.

(83) In some embodiments, the process **700** may include a step **703** in which the transceiver **101** conveys the first analyte information. In some embodiments, the first analyte information may be conveyed wirelessly. In some embodiments, the transceiver **101** may convey the first analyte information using the wireless communication IC **910**. In some embodiments, the transceiver **101** may employ a communication standard (e.g., a Bluetooth™ LE standard) to convey the first analyte information. In some alternative embodiments, the first analyte information may be conveyed using a wired connection (e.g., using a wired connection between the connector **902** of the transceiver **101** and the connector **30** of the display device **105**).

(84) In some embodiments, the process **700** may include a step **704** in which the display device **105** receives the first analyte information conveyed by the transceiver **101**. In some embodiments, the display device **105** may use the first wireless communication IC **312** to receive the first analyte information conveyed by the transceiver **101**.

(85) In some embodiments, the process **700** may include a step **705** in which the display device **105** receives second sensor data directly from the analyte sensor **100**. In some non-limiting embodiments, in step **704**, the display device **105** may cause the sensor **100** to take one or more measurements and convey the second sensor data to the display device **105** (e.g., by conveying one or more measurement commands to the analyte sensor **100**). In some embodiments, during step **705**, the display device **105** may have been positioned such that the display device **105** is operatively linked to the analyte sensor **100**. In some embodiments, the display device **105** may be operatively linked to the analyte sensor **100** by positioning the display device **105** in proximity to the analyte sensor **100** (e.g., by hovering or swiping the display device **105** in proximity to the analyte sensor **100**). In some non-limiting embodiments, during step **705**, the transceiver **101** is not positioned in proximity to the analyte sensor **100**, and/or the transceiver **101** is not operatively linked to the analyte sensor **100**. In some embodiments, the display device **105** may use the third

wireless communication IC **318** to receive the second sensor data directly from the analyte sensor **100**. In some embodiments, the display device **105** may receive the second sensor data from the analyte sensor **100** wirelessly using a communication standard such as, for example and without limitation, an NFC standard. In some non-limiting embodiments, the analyte sensor **101** may convey the second sensor data by modulating an electromagnetic wave generated by the display device **105**, and the display device **105** receiving the second sensor data may include detecting the modulations. In some non-limiting embodiments, the analyte sensor **100** may use the electromagnetic wave to power the analyte sensor **100**. In some embodiments, the second sensor data may include one or more raw measurements (e.g., one or more light measurements and one or more temperature measurements).

(86) In some embodiments, the process **700** may include a step **706** in which the display device **105** calculates second analyte information using at least the second sensor data. In some embodiments, the calculated second analyte information may include calculating one or more of an analyte concentration and an analyte concentration trend. In some embodiments, step **706** may include the display device **105** determining whether an alert and/or alarm condition exists based on the second analyte information (e.g., a calculated analyte concentration and/or analyte concentration trend). In some embodiments, the alert and/or alarm conditions may include one or more of a high analyte alarm level, a low analyte alarm level, a high target analyte level, and a low target analyte level.

(87) In some embodiments, the process **700** may include a step **707** in which the display device **105** displays the first analyte information and the second analyte information. In some embodiments, the MMA being executed by the computer **310** of the display device **105** may cause the user interface **340** of the display device **105** to display the first and second analyte information. For example, in some embodiments, the display device **105** may display the first analyte information after it is received from the transceiver **101** and then additionally display the second analyte information after it is calculated by the display device **105**. In some embodiments, step **707** may include displaying one or more of glucose data, trends, graphs, alarms, and alerts.

(88) In some embodiments, the process **700** may include an additional step in which the display device **105** may convey the first analyte information and the second analyte information over a network to a remote device. In some embodiments, the step of conveying the first analyte information and the second analyte information may include the display device **105** using the second wireless communication IC **316** to transmit the first and second sets of analyte information via network, such as for example and without limitation, a wireless local area network (e.g., Wi-Fi), a cellular network, and/or the Internet.

(89) While the process **700** described above is shown in a sequence of steps, it should be understood that the sequence of steps may be altered, additional steps may be added, and some steps may be omitted without departing from the scope of the present disclosure. For example, in some alternative embodiments, steps **705** and **706** may be performed before steps **701-704**. For another example, in some embodiments, process **700** may include additional steps in which the transceiver **101** receives third sensor data from the analyte sensor **100**, the transceiver **101** calculates third analyte information using at least the third sensor data, the transceiver **101** conveys the third analyte information to the display device, and the display device **105** displays the first, second, and third analyte information. For yet another example, the process **700** may include additional steps in which the display device **105**, receives fourth sensor data directly from the analyte sensor **100**, calculates fourth analyte information, and displays at least the first, second, and fourth analyte information.

(90) FIG. **10** is a flow chart illustrating an alternative process **800** embodying aspects of the present invention. The process **800** is similar to the process **700** described above with reference to FIG. **9** except that, while the analyte monitoring system **50** is being used in accordance with the CGM mode, the display device **105** calculates analyte information using sensor data received indirectly from the analyte sensor **100** via the transceiver **101** (instead of receiving from the transceiver **101**

analyte information calculated by the transceiver **101** as in step **703** of the process **700**). In some embodiments, the process **800** may include one or more steps (e.g., steps **801-803**) in which the analyte monitoring system **50** is used according to the CGM mode and one or more steps (e.g., steps **804-805**) in which the analyte monitoring system **50** is used according to the FGM mode.

(91) In some embodiments, the process **800** may include a step **801** in which the transceiver **101** receives first sensor data directly from an analyte sensor **100**. See description of step **701** above.

(92) In some embodiments, the process **800** may include a step **802** in which the transceiver **101** conveys the first sensor data. In some embodiments, the first sensor data may be conveyed wirelessly. In some embodiments, the transceiver **101** may convey the first sensor data using the wireless communication IC **910**. In some embodiments, the transceiver **101** may employ a communication standard (e.g., a Bluetooth™ LE standard) to convey the first sensor data. In some alternative embodiments, the first sensor data may be conveyed using a wired connection (e.g., using a wired connection between the connector **902** of the transceiver **101** and the connector **30** of the display device **105**).

(93) In some embodiments, the process **800** may include a step **803** in which the display device **105** receives the first sensor data conveyed by the transceiver **101**. In some embodiments, the display device **105** may use the first wireless communication IC **312** to receive the first sensor data conveyed by the transceiver **101**.

(94) In some embodiments, the process **800** may include a step **804** in which the display device **105** calculates first analyte information using at least the received first sensor data. In some embodiments, the calculated first analyte information may include one or more of an analyte concentration and an analyte concentration trend. In some embodiments, the step **804** of calculating the first analyte information may include determining if an alert and/or alarm condition exists based on at least the calculated first analyte information (e.g., an analyte concentration and/or analyte concentration trend). In some embodiments, the alert and/or alarm conditions may include one or more of a high analyte alarm level, a low analyte alarm level, a high target analyte level, and a low target analyte level.

(95) In some embodiments, the process **800** may include a step **805** in which the display device **105** to receive second sensor data directly from the analyte sensor **100**. See the description of step **705** above.

(96) In some embodiments, the process **800** may include a step **806** in which the display device **105** calculates second analyte information using at least the second sensor data. See description of step **706** above.

(97) In some embodiments, the process **800** may include a step **807** in which the display device **105** displays the first analyte information and the second analyte information. See description of step **707** above.

(98) In some embodiments, the process **800** may include an additional step in which the display device **105** may convey the first analyte information and the second analyte information over a network to a remote device. In some embodiments, the step of conveying the first analyte information and the second analyte information may include the display device **105** using the second wireless communication IC **316** to transmit the first and second sets of analyte information via network, such as for example and without limitation, a wireless local area network (e.g., Wi-Fi), a cellular network, and/or the Internet.

(99) While the process **800** described above is shown in FIG. **10** as a sequence of steps, it should be understood that the sequence of steps may be altered, additional steps may be added, and some steps may be omitted without departing from the scope of the present disclosure. For example, in some alternative embodiments, steps **805** and **806** may be performed before steps **801-804**. For another example, in some embodiments, process **800** may include additional steps in which the transceiver **101** receives third sensor data from the analyte sensor **100**, the transceiver **101** conveys the third sensor data to the display device **105**, the display device **105** calculates third analyte

information using at least the third sensor data, and the display device **105** displays the first, second, and third analyte information. For yet another example, the process **800** may include additional steps in which the display device **105**, receives fourth sensor data directly from the analyte sensor **100**, calculates fourth analyte information, and displays at least the first, second, and fourth analyte information.

(100) FIG. **11** is a flow chart illustrating a process **1100**, which may be executed by a computer **310** in a display device **105**, embodying aspects of the present invention. In some embodiments, the process **1100** may include one or more steps (e.g., steps **1101-1102**) in which the analyte monitoring system **50** is used according to the CGM mode and one or more steps (e.g., steps **1103-1105**) in which the analyte monitoring system **50** is used according to the FGM mode.

(101) In some embodiments, the process **1100** may include a step **1101** in which the display device **105** (e.g., the computer **310** of the display device **105**) determines whether the display device **105** has received analyte information conveyed by the transceiver **101**. In some embodiments, the analyte information may include one or more of: (i) an analyte concentration, (ii) a time stamp, and (iii) analyte concentration trend information. In some embodiments, the analyte information may additionally or alternatively include one or more of: (i) an alert, (ii) an alarm, and (iii) a notification. In some embodiments, if analyte information has been received from the transceiver **101**, the process **1100** may proceed from step **1101** to a step **1102** in which the display device **105** stores the analyte information (e.g., in the memory **314** of the display device **105**) and/or displays the analyte information (e.g., using one or more of a display **320**, a speaker **324**, and a vibration motor **326** of a user interface **340** of the display device **105**). In some embodiments, if analyte information from the transceiver **101** has not been received by the display device **105**, the process **1100** may proceed from the step **1101** to a step **1103**.

(102) In some embodiments, the process **1100** may include a step **1103** in which the display device **105** (e.g., the computer **310** of the display device **105**) determines whether the display device **105** has received sensor data directly from the analyte sensor **100**. In some embodiments, the sensor data may include one or more of: (i) a measurement of one or more detectable properties of the analyte indicator **106** and (ii) a temperature measurement. In some embodiments, if the display device **105** has received sensor data directly from the analyte sensor **100**, the process **1100** may proceed to a step **1104** in which the display device **105** (e.g., the computer **310** of the display device **105**) calculates analyte information using at least the sensor data received directly from the analyte sensor **100** (see description of step **705** of FIG. **7** above) and then to a step **1105** in which the display device **105** stores the analyte information (e.g., in a memory **314** of the display device **105**) and/or displays the analyte information (e.g., using one or more of a display **320**, a speaker **324**, and a vibration motor **326** of a user interface **340** of the display device **105**). In some embodiments, as illustrated in FIG. **11**, the process **1100** may proceed from step **1105** back to step **1101**. However, this is not required, and, in some alternative embodiments, the process **1100** may proceed from step **1105** to a step **1106**. In some embodiments, if the display device **105** has not received sensor data directly from an analyte sensor **100**, the process **1100** may proceed from the step **1103** to a step **1106**.

(103) In some embodiments, the step **1104** of calculating analyte information may include calculating an analyte concentration and/or an analyte concentration trend using at least the received sensor data. In some embodiments, the step **1104** of calculating the analyte information includes determining if an alert and/or alarm condition exists based on the calculated analyte concentration and/or analyte concentration trend. In some embodiments, the alert and/or alarm conditions include one or more of a high analyte alarm level, a low analyte alarm level, a high target analyte level, and a low target analyte level.

(104) In some embodiments, the process **1100** may include a step **1106** in which the display device **105** (e.g., the computer **310** of the display device **105**) determines whether one or more calibration points have been received by the user interface **340** of the display device **105** (e.g., by the user

input **322** of the user interface **340**). In some embodiments, the one or more calibration points may include one or more reference measurements (e.g., one or more self-monitoring blood glucose (SMBG) measurements). In some embodiments, if one or more calibration points have been received by the user interface **340** of the display device **105**, the process **1100** may proceed from the step **1106** to a step **1107** in which the display device **105** performs a calibration of conversion of sensor data to analyte information using at least the one or more calibration points and a step **1108** in which the display device **105** conveys the one or more calibration points to the transceiver **101**. In some embodiments, if one or more calibration points have not been received by the display device **105**, the process **1100** may from the step **1106** back to step **1101**.

(105) FIG. **12** is a flow chart illustrating a process **1200**, which may be executed by a computer **310** in a display device **105**, embodying aspects of the present invention. The process **1200** is similar to the process **1100** described above with reference to FIG. **11** except that, while the analyte monitoring system **50** is being used in accordance with the CGM mode, the display device **105** calculates analyte information using sensor data received indirectly from the analyte sensor **100** via the transceiver **101** (instead of receiving from the transceiver **101** analyte information calculated by the transceiver **101** as in step **1101** of process **1100**). In some embodiments, the process **1200** may include one or more steps (e.g., step **1201**) in which the analyte monitoring system **50** is used according to the CGM mode and one or more steps (e.g., step **1104**) in which the analyte monitoring system **50** is used according to the FGM mode.

(106) In some embodiments, the process **1200** may include a step **1201** in which the display device **105** (e.g., the computer **310** of the display device **105**) determines whether the display device **105** has received sensor data conveyed by the transceiver **101**. In some embodiments, if the display device **105** has received sensor data from the transceiver **101**, the process **1200** may proceed to a step **1202** of calculating analyte information and a step **1203** of storing and/or displaying the analyte information. In some embodiments, if the display device **105** has not received sensor data from the transceiver **101**, the process **1200** may proceed from the step **1201** to a step **1204**.

(107) In some embodiments, the process **1200** may include a step **1204** of determining whether the display device **105** has received sensor data directly from the analyte sensor **100**. In some embodiments, if the display device **105** has received sensor data directly from the analyte sensor **100**, the process **1200** may proceed from the step **1204** to the step **1202** of calculating analyte information and the step **1203** of storing and/or displaying the analyte information. In some embodiments, if the display device **105** has not received sensor data directly from the analyte sensor **100**, the process **1200** may proceed from the step **1204** to a step **1205**.

(108) In some embodiments, the process **1200** may include a step **1202** in which the display device **105** (e.g., the computer **310** of the display device **105**) calculates analyte information using at least the sensor data received (a) indirectly from the analyte sensor **100** via the transceiver **101** in the step **1201** or (b) directly from the analyte sensor **100** in step **1204**. In some embodiments, calculating the analyte information in step **1202** may include calculating an analyte concentration and/or an analyte concentration trend using at least the received sensor data. In some embodiments, the step **1202** of calculating the analyte information may include determining if an alert and/or alarm condition exists based on the calculated analyte concentration and/or the analyte concentration trend. In some embodiments, the alert and/or alarm conditions include one or more of a high analyte alarm level, a low analyte alarm level, a high target analyte level, and a low target analyte level. In some embodiments, the process **1200** may proceed from the step **1202** to a step **1203**.

(109) In some embodiments, the process **1200** may include a step **1203** in which the display device **105** stores the analyte information (e.g., in a memory **314** of the display device **105**) and/or displaying the analyte information (e.g., using one or more of a display **320**, a speaker **324**, and a vibration motor **326** of a user interface **340** of the display device **105**). In some embodiments, as shown in FIG. **12**, the process **1200** may proceed from step **1203** back to the step **1201**. However,

this is not required, and, in some alternative embodiments, the process **1200** may proceed from step **1203** to the step **1205**.

(110) In some embodiments, the process **1200** may include a step **1205** in which the display device **105** (e.g., the computer **310** of the display device **105**) determines whether one or more calibration points have been received by the user interface **340** of the display device **105** (e.g., by the user input **322** of the user interface **340**). In some embodiments, the one or more calibration points include one or more reference measurements (e.g., one or more self-monitoring blood glucose (SMBG) measurements). In some embodiments, if the display device **105** has received one or more calibration points, the process **1200** may proceed from the step **1205** to a step **1206** in which the display device **105** performs a calibration of the conversion of sensor data to analyte information using at least the one or more calibration points and a step **1207** in which the display device **105** conveys the one or more calibration points to the transceiver **101**. However, in some alternative embodiments, the process **1200** may not include the step **1207**, and the process **1200** may proceed from step **1206** back to step **1201**. In some of these alternative embodiments, the transceiver **101** may not calculate analyte information and, therefore, would not require the one or more calibration points. In some embodiments, if one or more calibration points have not been received by the display device **105**, the process **1200** may proceed from step **1205** back to step **1201**.

(111) While the processes **1100** and **1200** are illustrated in FIGS. **11** and **12**, respectively, as sequences of steps, it should be understood that the sequence of steps may be altered, additional steps may be added, and some steps may be omitted without departing from the scope of the present disclosure. For example, the steps of processes **1100** and **1200** may be performed in a different order.

(112) While the subject matter of this disclosure has been described and shown in considerable detail with reference to certain illustrative embodiments, including various combinations and sub-combinations of features, those skilled in the art will readily appreciate other embodiments and variations and modifications thereof as encompassed within the scope of the present disclosure. Moreover, the descriptions of such embodiments, combinations, and sub-combinations is not intended to convey that the claimed subject matter requires features or combinations of features other than those expressly recited in the claims. Accordingly, the scope of this disclosure is intended to include all modifications and variations encompassed within the spirit and scope of the following appended claims.

Claims

1. An analyte monitoring system comprising: an analyte sensor including an analyte indicator that exhibits one or more detectable properties based on an amount or concentration of an analyte in proximity to the analyte indicator; a transceiver configured to: receive first sensor data directly from the analyte sensor, generate first analyte information using at least the received first sensor data, wherein generating the first analyte information comprises calculating a first analyte concentration using at least the received first sensor data, and convey the first analyte information; and a display device configured to: receive second sensor data directly from the analyte sensor, generate second analyte information using at least the received second sensor data, wherein generating the second analyte information comprises calculating a second analyte concentration using at least the received second sensor data, receive the first analyte information conveyed by the transceiver, and display the first and second analyte concentrations.
2. The analyte monitoring system of claim 1, wherein the display device comprises a first wireless communication integrated-circuit (IC) and a second wireless communication IC, the first communication IC is configured to employ a first standard to communicate wirelessly, the second communication IC is configured to employ a second standard to communicate wirelessly, and the second standard is different than the first standard.

3. The analyte monitoring system of claim 2, wherein the first standard is a Bluetooth standard, and the second standard is a near field communication (NFC) standard.
4. The analyte monitoring system of claim 2, wherein the display device is configured to use the first wireless communication IC to receive the first analyte information conveyed by the transceiver and to use the second wireless communication IC to receive the second sensor data directly from the analyte sensor.
5. The analyte monitoring system of claim 2, wherein the display device comprises a third wireless communication IC, and the display device is configured to use the third wireless communication IC to convey the first and second analyte information over a network to a remote device.
6. The analyte monitoring system of claim 1, wherein the transceiver is configured to be worn by a host using the analyte sensor while the transceiver receives the first sensor data directly from the analyte sensor.
7. The analyte monitoring system of claim 1, wherein one or more of the first and second analyte information comprises one or more of: (i) a time stamp and (ii) analyte concentration trend information.
8. The analyte monitoring system of claim 1, wherein one or more of the first and second analyte information comprises one or more of: (i) an alert, (ii) an alarm, and (iii) a notification.
9. The analyte monitoring system of claim 1, wherein the first sensor data comprises one or more of: (i) a measurement of the one or more detectable properties and (ii) a temperature measurement.
10. The analyte monitoring system of claim 1, wherein the display device comprises a graphical user interface and is configured to generate an alert or alarm on the graphical user interface of the display device.
11. The analyte monitoring system of claim 1, wherein the transceiver comprises a graphical user interface and is configured to generate an alert or alarm on the graphical user interface of the transceiver.
12. The analyte monitoring system of claim 1, wherein the display device is configured to convey the second analyte information to the transceiver, and the transceiver is configured to receive the second analyte information.
13. The analyte monitoring system of claim 1, wherein the display device is configured to receive one or more calibration points.
14. The analyte monitoring system of claim 13, wherein the display device is configured to perform an analyte information calibration using at least the one or more calibration points.
15. The analyte monitoring system of claim 13, wherein the display device is configured convey the one or more calibration points, and the transceiver is configured to receive the one or more calibration points and perform an analyte information calibration using at least the one or more calibration points.
16. A method comprising: (a) using a transceiver of an analyte monitoring system to receive first sensor data directly from an analyte sensor of the analyte monitoring system; (b) using the transceiver to generate first analyte information using at least the first sensor data, wherein generating the first analyte information comprises calculating a first analyte concentration using at least the received first sensor data; (c) using the transceiver to convey the first analyte information; (d) using a display device of the analyte monitoring system to receive the first analyte information conveyed by the transceiver; (e) using the display device to receive second sensor data directly from the analyte sensor; (f) using the display device to generate second analyte information using at least the received second sensor data, wherein generating the second analyte information comprises calculating a second analyte concentration using at least the received second sensor data; and (g) using the display device to display the first analyte information and the second analyte concentrations.
17. The method of claim 16, wherein the step (a) comprises positioning the transceiver such that the transceiver is operatively linked to the analyte sensor.

18. The method of claim 17, wherein positioning the transceiver includes wearing, by a host using the analyte sensor, the transceiver externally on an armband, a wrist band, a waist band, or an adhesive patch.
 19. The method of claim 16, wherein the step (e) comprises positioning the display device such that the display device is operatively linked to analyte sensor.
 20. The method of claim 16, wherein the display device comprises a first wireless communication IC and a second wireless communication IC, the first communication IC is configured to employ a first standard to communicate wirelessly, the second communication IC is configured to employ a second standard to communicate wirelessly, and the second standard is different than the first standard.
 21. The method of claim 20, wherein the first standard is a Bluetooth standard, and the second standard is a near field communication (NFC) standard.
 22. The method of claim 20, wherein the step (c) comprises using the first communication IC of the display device to receive the first analyte information conveyed by the transceiver.
 23. The method of claim 20, wherein the step (e) comprises using the second communication IC of the display device to receive the second sensor data directly from the analyte sensor.
 24. The method of claim 20, wherein the display device comprises a third wireless communication IC, and the method further comprises: (h) using the third wireless communication of the display device to convey the first analyte information and the second analyte information over a network to a remote device.
 25. The method of claim 16, wherein the step (e) comprises removing the transceiver away from the analyte sensor such that the transceiver is not operatively linked to the analyte sensor.
 26. The method of claim 16, wherein one or more of the first and second analyte information comprises one or more of: (i) a time stamp and (ii) analyte concentration trend information.
 27. The method of claim 16, wherein one or more of the first and second analyte information comprises one or more of: (i) an alert, (ii) an alarm, and (iii) a notification.
 28. The method of claim 16, wherein the first sensor data comprises one or more of: (i) a measurement of one or more detectable properties exhibited by an analyte indicator of the analyte sensor based on an amount or concentration of an analyte in proximity to the analyte indicator and (ii) a temperature measurement.
 29. The method of claim 16, wherein the display device comprises a graphical user interface and is configured to generate an alert or alarm on the graphical user interface of the display device.
 30. The method of claim 16, wherein the transceiver comprises a graphical user interface and is configured to generate an alert or alarm on the graphical user interface of the transceiver.
 31. The method of claim 16, further comprising: using the display device to convey the second analyte information; and using the transceiver to receive the second analyte information conveyed by the display device.
 32. The method of claim 16 further comprising using the display device to receive one or more calibration points.
 33. The method of claim 32, further comprising using the display device to perform an analyte information calibration using at least the one or more calibration points.
 34. The method of claim 32 further comprising: using the display device to convey the one or more calibration points; and using the transceiver to receive the one or more calibration points conveyed by the display device and perform an analyte information calibration using at least the one or more calibration points.
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